+3V3 rail

GND rail

① POWER 5V→AMS1117-3.3→3V3

J12 DC 5V IN
1 +5V
2 GND
screw terminal

U1 AMS1117-3.3
+5V
+3V3
GND
LDO SOT-223

C1 10uF/25V
1 +5V
2 GND
input cap

C2 22uF
1 +3V3
2 GND
output cap

R4 1K 0603
1 +3
2 NL
LED limit

LED1 LED GREEN
A NL
K GN
power led

② MCU ESP32-S3-DevKitC-1 (plug onto J1)

J1 ESP32-S3-DevKitC-1
1 +3V3
2 GND
3 NEN
4 NB0
5 MIC_SCK
6 MIC_WS
7 MIC_SD
8 SPK_DIN
9 SPK_BCLK
10 SPK_LRC
11 SDA
12 SCL
13 SERVO_PAN
14 SERVO_TILT
15 LDR_ADC
16 TOUCH
17 NLED
18 UTX0
19 URX0
插板主头(功能序)

③ AUDIO MIC+AMP

J3 INMP441 MIC
VDD
GND
L/R
SCK
WS
SD
1x6 2.54

C3 100nF
1 +3V3
2 GND
mic decoupling

J4 MAX98357A AMP
VIN
GND
SD
GAIN
DIN
BCLK
LRC
OUT+
OUT-
1x9 2.54

J5 SPEAKER 8R3W
NC_SPK 1
NC_SPK 2
2P connector

④ DISPLAY

J6 OLED SSD1306
VCC
GND
SCL
SDA
1x4 2.54

R2 4.7K 0603
1 +3V3
2 NC_SCL
I2C pullup

R3 4.7K 0603
1 +3V3
2 NC_SDA
I2C pullup

⑤ SENSOR

J7 LDR GL5528
+3V3 TOP
NLDR BOT

J8 TOUCH FOIL
T
0 TP

R1 10K 0603
1 NLDR
2 GND

⑥ SERVO external 5V only!

J15 SERVO 5V IN
+5V_SERVO 1
GND 2
screw terminal

J9 SG90 pan
NSP SIG
+5V_SERVO VCC
GND GND

J10 SG90 tilt
NST SIG
+5V_SERVO VCC
GND GND

SW2 RST(EN)
1 NEN
2 GND
tact

R5 10K 0603
1 NEN
2 +3V3
EN pullup

SW1 BOOT(IO0)
1 NB0
2 GND
tact

R6 10K 0603
1 NB0
2 +3V3
IO0 pullup

J14 UART PROG
1 +3V3
2 UTX0
3 URX0
4 NEN
5 NB0
6 GND
1x6 2.54 header